UNIFORM HIGH-TWIST VANISHING AND SECTION LIFTING FROM AN AMPLE SCHEME FIBRE OF A PROPER ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. The conclusion concerns stalks aty for alld>=d_0. The fibreX_y is expressly assumed to be a scheme; no global scheme assumption onX is introduced.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Proper algebraic spaces over a Noetherian scheme, coherent sheaves, ampleness andSerre vanishing on the chosen scheme fibre, finite graded modules, flat base change and the theorem on formal functions.

Source reference. Original equation 02NA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3473–3473. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 073K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9064–9064. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 08AU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10544–10544. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08AM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10537–10537. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0897: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10487–10487. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D2U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16394–16394. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:spaces-divisors.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original theorem 08AZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10549–10549. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08AS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10542–10542. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08AY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10548–10548. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let Y be a Noetherian scheme. Let X be an algebraic space over Y such that the structure morphism $f : X \to Y$ is proper. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Let $y \in Y$ be a point such that X_y is a scheme and $\mathcal{L}_y$ is ample on X_y. Then there exists a d_0 such that for all $d \geq d_0$ we have*

$$R^p f_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d})_y = 0 \text{ for } p > 0$$

and the map

$$f_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d})_y \longrightarrow H^0(X_y, \mathcal{F}_y \otimes_{\mathcal{O}_{X_y}} \mathcal{L}_y^{\otimes d})$$

is surjective.

Proof. Note that $\mathcal{O}_{Y,y}$ is a Noetherian local ring. Consider the canonical morphism $c : \mathrm{Spec}(\mathcal{O}_{Y,y}) \to Y$, see Schemes, Equation (02NA). This is a flat morphism as it identifies local rings. Denote momentarily $f' : X' \to \mathrm{Spec}(\mathcal{O}_{Y,y})$ the base change of f to this local ring. We see that $c^* R^p f_* \mathcal{F} = R^p f'_* \mathcal{F}'$ by Cohomology of Spaces, Lemma 073K. Moreover, the fibres X_y and X'_y are identified. Hence we may assume that $Y = \mathrm{Spec}(A)$ is the spectrum of a Noetherian local ring $(A, \mathfrak{m}, \kappa)$ and $y \in Y$ corresponds to $\mathfrak{m}$. In this case $R^p f_*(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d})_y = H^p(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d})$ for all $p \geq 0$. Denote $f_y : X_y \to \mathrm{Spec}(\kappa)$ the projection.

Let $B = \mathrm{Gr}_{\mathfrak{m}}(A) = \bigoplus_{n \geq 0} \mathfrak{m}^n/\mathfrak{m}^{n+1}$. Consider the sheaf $\mathcal{B} = f_y^* \widetilde{B}$ of quasi-coherent graded $\mathcal{O}_{X_y}$-algebras. We will use notation as in Cohomology of Spaces, Section 08AU with I replaced by $\mathfrak{m}$. Since X_y is the closed subspace of X cut out by $\mathfrak{m}\mathcal{O}_X$ we may think of $\mathfrak{m}^n\mathcal{F}/\mathfrak{m}^{n+1}\mathcal{F}$ as a coherent $\mathcal{O}_{X_y}$-module, see Cohomology of Spaces, Lemma 08AM. Then $\bigoplus_{n \geq 0} \mathfrak{m}^n\mathcal{F}/\mathfrak{m}^{n+1}\mathcal{F}$ is a quasi-coherent graded $\mathcal{B}$-module of finite type because it is generated in degree zero over $\mathcal{B}$ abd because the degree zero part is $\mathcal{F}_y = \mathcal{F}/\mathfrak{m}\mathcal{F}$ which is a coherent $\mathcal{O}_{X_y}$-module. Hence by Cohomology of Schemes, Lemma 0897 part (2) there exists a d_0 such that

$$H^p(X_y, \mathfrak{m}^n\mathcal{F}/\mathfrak{m}^{n+1}\mathcal{F} \otimes_{\mathcal{O}_{X_y}} \mathcal{L}_y^{\otimes d}) = 0$$

for all $p > 0$, $d \geq d_0$, and $n \geq 0$. By Cohomology of Spaces, Lemma 0D2U this is the same as the statement that $H^p(X, \mathfrak{m}^n\mathcal{F}/\mathfrak{m}^{n+1}\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d}) = 0$ for all $p > 0$, $d \geq d_0$, and $n \geq 0$.

Consider the short exact sequences

$$0 \to \mathfrak{m}^n\mathcal{F}/\mathfrak{m}^{n+1}\mathcal{F} \to \mathcal{F}/\mathfrak{m}^{n+1}\mathcal{F} \to \mathcal{F}/\mathfrak{m}^n\mathcal{F} \to 0$$

of coherent $\mathcal{O}_X$-modules. Tensoring with $\mathcal{L}^{\otimes d}$ is an exact functor and we obtain short exact sequences

$$0 \to \mathfrak{m}^n\mathcal{F}/\mathfrak{m}^{n+1}\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d} \to \mathcal{F}/\mathfrak{m}^{n+1}\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d} \to \mathcal{F}/\mathfrak{m}^n\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d} \to 0$$

Using the long exact cohomology sequence and the vanishing above we conclude (using induction) that

(1) $H^p(X, \mathcal{F}/\mathfrak{m}^n\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d}) = 0$ for all $p > 0$, $d \geq d_0$, and $n \geq 0$, and
(2) $H^0(X, \mathcal{F}/\mathfrak{m}^n\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d}) \to H^0(X_y, \mathcal{F}_y \otimes_{\mathcal{O}_{X_y}} \mathcal{L}_y^{\otimes d})$ is surjective for all $d \geq d_0$ and $n \geq 1$.

By the theorem on formal functions (Cohomology of Spaces, Theorem 08AZ) we find that the $\mathfrak{m}$-adic completion of $H^p(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d})$ is zero for all $d \geq d_0$ and $p > 0$. Since $H^p(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d})$ is a finite A-module by Cohomology of Spaces, Lemma 08AS it follows from Nakayama's lemma (Algebra, Lemma 00DV) that $H^p(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d})$ is zero for all $d \geq d_0$ and $p > 0$. For $p = 0$ we deduce from Cohomology of Spaces, Lemma 08AY part (3) that $H^0(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d}) \to H^0(X_y, \mathcal{F}_y \otimes_{\mathcal{O}_{X_y}} \mathcal{L}_y^{\otimes d})$ is surjective, which gives the final statement of the lemma. □